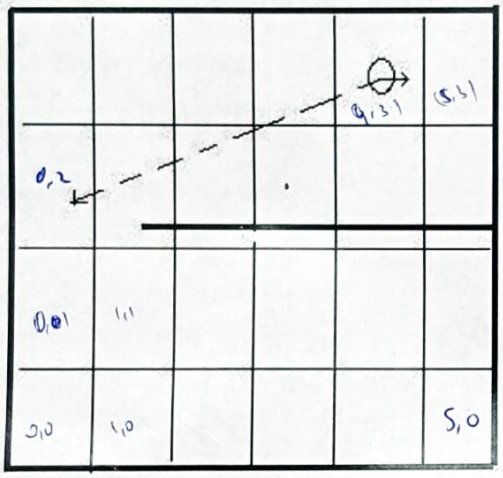
Intro to Robotics: Perception and Planning

Spring 2026

1. Figure below shows a map discretized into cells. The obstacles are shown by dark lines. The robot is in one of the cells and is shown as a circle. The robot points along 0 degrees (along X axis, shown by an arrow towards the right) and is aware of the orientation.



a). What is the initial belief at each cell ? **(1 point)**

b). Define a simple sensor update function, i.e, P(Z/X) without involving exponential and other such complex terms. Assume that the robot has four sensors seeing right, top, left and bottom. (Hint: The sensor update function can be defined in terms of cell distances than absolute distances.) **(2 points)**

c). What are the belief values after performing a sensor update? Clearly write the equations involved and how each term in the equation is getting plugged with values **(3 points)**

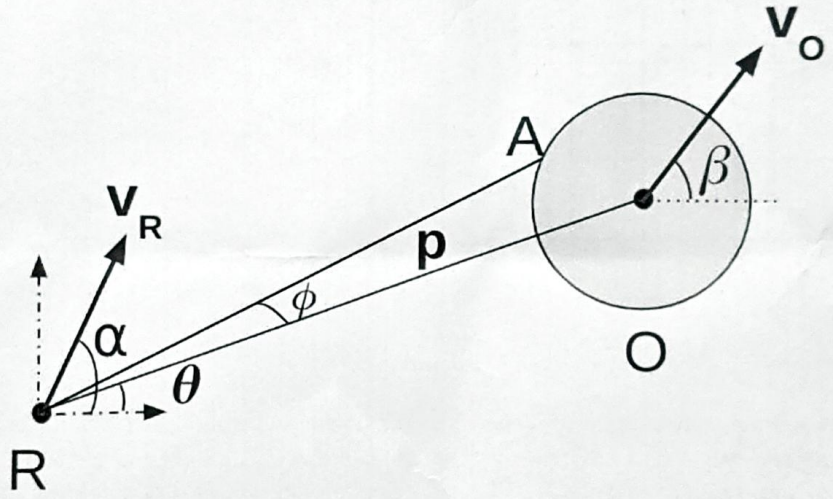
d). The robot then moves to the cell shown by the arrow. What is the update equation that governs the belief of the cells after a motion **(1 point)**

e) What are the belief values at the cells after the motion update is performed? **(3 points)**

2. Consider a point robot R and a circular object O, as illustrated in the figure below, under the following initial conditions.:

- Relative distance between them: $|\mathbf{p}| = 10$ units
- Orientation of $\mathbf{p}$, $\boldsymbol{\theta} = 35^0$
- Orientation of $\mathbf{v_O}$, $\beta = 50^0$
- $\mathbf{v_R} = 6$ m/s
- $\mathbf{v_O} = 4.5$ m/s
- Object radius = 3
- For a point A on O, $\phi = 10^0$

Find the value of α for which **R** is on a collision course with the point A. Explain how you arrive at the solution. **(10 points)**



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